PEER REVIEW HISTORY

BMJ Open publishes all reviews undertaken for accepted manuscripts. Reviewers are asked to complete a checklist review form (**http://bmjopen.bmj.com/site/about/resources/checklist.pdf**) and are provided with free text boxes to elaborate on their assessment. These free text comments are reproduced below.

ARTICLE DETAILS

TITLE (PROVISIONAL)	Checklists to Reduce Diagnostic Error: A Systematic Review of the Literature Using a Human Factors Framework
AUTHORS	Al-Khafaji, Jawad; Townsend, Ryan; Townsend, Whitney; Chopra, Vineet; Gupta, Ashwin

VERSION 1 – REVIEW

REVIEWER	Ashley Meyer Baylor College of Medicine
REVIEW RETURNED	05-Nov-2021

GENERAL COMMENTS	Preventing diagnostic error is an important endeavor and very few interventions have been shown to be effective to date. It is helpful to have this review show the culmination of evidence thus far for checklists- an intervention that could be easily scalable and has been used effectively in other domains. I enjoyed reading the article, but think some tweaks could make it more suitable for publication. Minor: Abstract: it would be helpful for readers to know which diagnostic outcomes you are referring to (e.g., diagnostic accuracy, timeliness, etc.). Abstract: I'm not sure if it could be worked in given word count constraints, but a general summary of the quality of the articles could be useful for readers up front. Discussion: how do these findings compare to other reviews of diagnostic interventions that have been published- for example, there are two older papers on cognitive and system interventions to improve diagnosis. Do these findings line up with those? Have a lot more checklists been examined since then? Putting this review into context of previous ones would be helpful for readers. Moderate: Rationale: the main thing that needs strengthening is the rationale for examining "whether or how human and systems factors are integrated into the checklists"- it is not strong as written in the introduction. It seems that the SEIPS 2.0 model serves as a helpful way to think about which factors are covered or are not covered by the checklists (since it's pretty comprehensive- it's a good way to see what has been covered or not), but beyond that it is not clear up front how figuring out which checklists cover which factors in the model pushes the science forward or tells people what should be done next. That being said, I think the finding that task-oriented checklists seem to be more effective than cognitive-based ones suggests that sorting the checklists into the things they are trying to change could be helpful for the development of future checklists (or if in a certain context it is known that a specific aspect of the SEIPS

	model has issues (e.g., in a certain clinic), a checklist aimed at improving that aspect could be very useful. As it reads, my main takeaway is that checklists that are task-oriented (which I'm thinking are likely more concrete in their guidance) are more effective than general cognitive ones (which I'm thinking are probably more vague in their guidance). Adding explicitly why examining the checklists along different aspects of the SEIPS model will improve the paper. Thanks for the opportunity to review and I hope these comments are helpful.

REVIEWER	Amy Z Q Chung Transport for New South Wales, Safety Environment and Regulation
REVIEW RETURNED	10-Dec-2021

GENERAL COMMENTS	Overall, this is an interesting and well-written paper. The topic itself is very practical and will hopefully contribute to better designed checklists in the medical setting to reduce diagnostic errors. I would like to see more detail in the Method section regarding the assessment of "risk of bias" to explain what this is, what it involves, and the categorisation used (e.g. good / poor quality), and a more detailed discussion in the Results section and what it means for this particular study. There is a "PRISMA 2020 for Abstracts Checklist" attached on the last page of this manuscript. This isn't referred to within the paper and it is unclear what purpose it serves. Is this the completed PRISMA checklist for this paper? If so, it does not make sense that the "risk of bias" under the "Method" section is reported as "No", because this paper does cover risk of bias. The reviewer provided a marked copy with additional comments. Please contact the publisher for full details.

VERSION 1 – AUTHOR RESPONSE

REVIEWER COMMENTS:

1- Please revise the formatting of your abstract so that it includes the following sections: Objectives >> Design >> Data Sources >> Eligibility Criteria >> Data extraction and synthesis >> Results >> Conclusions. Please see the following published abstract as an example: http://bmjopen.bmj.com/content/8/3/e019438

The abstract was revised to be consistent with the outlined structure (lines 52-79).

2- Can you please work on improving the conclusions section in the main manuscript? This section currently isn't very informative and reads more like a summary than a conclusions section.

We revised the conclusion section to emphasize the main takeaway points of the paper. The conclusion now reads as follows (lines 564-573):

"Checklists could hold great potential to reduce clinical diagnostic error; therefore, a comprehensive understanding of their landscape is crucial for further development. Organization of checklists by the SEIPS 2.0 is highly illustrative of the current state- highlighting areas of strength of current checklists as well as blind spots that remain poorly addressed. Generally, current published checklists emphasize very few SEIPS 2.0 elements, mainly the Task and Cognitive sub-components, while leaving others entirely un-utilized. Tasks-focused checklists seem to be more associated with a

significant reduction in diagnostic error. Understanding the impact of incorporating less explored SEIPS 2.0 subcomponents within checklists for diagnostic error warrants evaluation. Further, studies examining the ability of task-oriented checklists to reduce diagnostic error may ultimately facilitate improved diagnosis".

3- Along with your revised manuscript, please provide a completed copy of the PRISMA 2020 checklist (http://prisma-statement.org/prismastatement/Checklist.aspx). Please remember to include the relevant page number(s) from the manuscript next to each reporting item or state 'n/a' next to items that are not applicable to your study.

We have completed and included a PRISMA 2020 checklists for the abstract and the main text with the appropriate page number references.

4- It would be helpful to add more detail here about what the "risk of "bias" is referring to. Is this about the risk of bias when a user is applying a checklist, or risk of bias in the study that is examining one or more checklists? Please clarify.

In accordance with PRISMA guidelines, the risk of bias assessed in our paper refers to the risk of bias in an individual study evaluated by examining their designs, data synthesis and analysis, and outcome-measuring. We have edited the original text in the abstract and manuscript to better reflect that it is "risk of study bias" assessment. The abstract now reads "Risk of study bias was independently evaluated using standardized quality assessment tools in accordance with PRISMA" (lines 65-66). The methods verbiage has been updated to "Each study, when appropriate, was assessed for risk of bias (RoB) in its design, data synthesis and analysis, outcome-measuring, and conclusion, by two authors using the following established and validated quality assessment tools: Cochrane's RoB 2.0 tool for randomized studies,33 ROBINS-I tool for non-randomized quasi experimental studies,34 the Newcastle-Ottawa Scale for cohort studies (with control),35 and the NIH Quality Assessment tool for pre and posttest cohort studies without control (See Appendix for references to those tools)" (lines 251-257).

5- It would be useful to have a brief explanation here of what each of these levels (e.`g. low risk, good/poor quality) mean. Because risk of bias is determined separately for different types of studies - does each method to examine the risk of bias have a different categories of good vs. poor? It may be useful to include this information in the Method section.***
In order to assess the risk of bias (RoB) and the quality of the included studies, we used the following four validated quality assessment tools: Cochrane's RoB 2.0 tool for assessing randomized studies,3 ROBINS-I tool for assessing non-randomized quasi experimental studies,4 the Newcastle-Ottawa Scale for assessing cohort studies,5 and the NIH Quality Assessment tool for assessing pre and posttest cohort studies.6 We used these varying tools because they are validated for the study designs included within our review. The instruments are comprehensive and include multiple defined categories of risk and quality with regards to study design, data synthesis, and conclusion. Considering that it would take significant space to describe each tool in the manuscript, we have instead elected to reference the tools in the "Data Extraction and Quality Assessment" section in the Methods and have elaborated further on what they assess (lines 251-253, also see answer to item 4 above for verbiage used in the Methods session). In addition, we added a reference link to each of the four quality assessment tools in the Appendix (see Appendix-Page 7). The rating for each study is outlined in the Appendix-Table1. We also further elaborated on why three of the randomized-controlled trials were rated with "Some Concerns," and why three of the pre-posttest studies were rated with either 'Fair' or 'Poor) in the Discussion section, which now reads: "Indeed, when examining risk of bias, half of the RCTs (n=3) included in this review had 'Some Concerns', largely related to differences in baseline characteristics or limited analyses to estimate the effect of the checklist

intervention. In addition, three of the five pre-posttest studies were rated as either ‘Fair’ or ‘Poor’, largely due to lack of clear selection criteria of study population or the study sample being insufficient or not representative of the general population (Appendix-Table 1).” (lines 500-523).

Reviewer 1

6- Abstract: it would be helpful for readers to know which diagnostic outcomes you are referring to (e.g., diagnostic accuracy, timeliness, etc.).

Abstract was edited to read: “If reported, checklist efficacy in reducing diagnostic error (e.g., diagnostic accuracy, number of errors, or any patient-related outcomes)” in the Data Extraction and Synthesis section of the Abstract (lines 64-65).

7- Abstract: I'm not sure if it could be worked in given word count constraints, but a general summary of the quality of the articles could be useful for readers up front.

Due to the limitations of the word count, we are not able to include full details on quality of all the extracted articles. We have, instead, shown these in the Appendix-Table 1 where risk of study bias is tabulated clearly for those interested in these data, and included references to those tools in Appendix-Page 7. We have also discussed risk of study bias in the Discussion with regards to implications of our findings (as described in our answer to item 5).

8- Discussion: how do these findings compare to other reviews of diagnostic interventions that have been published- for example, there are two older papers on cognitive and system interventions to improve diagnosis. Do these findings line up with those? Have a lot more checklists been examined since then? Putting this review into context of previous ones would be helpful for readers.

We further elaborated on the clinical context of our review in the Discussion section, and how it compares -particularly in terms of published checklists- to prior literature describing cognitive and systems intervention, and included references to two prior review articles in this field (lines 385-389).7,8 Since these two original reviews were published, there has been a substantial increase in studies of checklists seeking to address diagnostic error. Our study also goes beyond simply reporting the current literature, and instead describes checklists through the lens of SEIPS 2.0 human factors framework to assess how these characteristics may influence diagnostic error. Therefore, we point out that our work is not simply a summary of what’s new, but rather an assessment of the literature in terms of what “may work” when it comes to reducing diagnostic errors.
This part of the Discussion section now reads: “Prior reviews on interventions intended to reduce diagnostic error describe cognitive and system-related interventions, but included small numbers of published clinical diagnosis checklists in this field.68,69 Since then, a plethora of additional studies targeting diagnostic error using checklists have been published. Despite this fact, the importance of checklists within the realm of diagnostic error remains unclear. In this study, we included 30 studies and 25 unique checklists, characterized each checklist based on SEIPS 2.0 framework, and assessed which human factors may have a role in influencing diagnostic errors” (lines 464-485).

9- Rationale: the main thing that needs strengthening is the rationale for examining "whether or how human and systems factors are integrated into the checklists"- it is not strong as written in the introduction. It seems that the SEIPS 2.0 model serves as a helpful way to think about which factors are covered or are not covered by the checklists (since it's pretty comprehensive- it's a good way to see what has been covered or not), but beyond that it is not clear up front how figuring out which

checklists cover which factors in the model pushes the science forward or tells people what should be done next. That being said, I think the finding that task-oriented checklists seem to be more effective than cognitive-based ones suggests that sorting the checklists into the things they are trying to change could be helpful for the development of future checklists (or if in a certain context it is known that a specific aspect of the SEIPS model has issues (e.g., in a certain clinic), a checklist aimed at improving that aspect could be very useful. As it reads, my main takeaway is that checklists that are task-oriented (which I'm thinking are likely more concrete in their guidance) are more effective than general cognitive ones (which I'm thinking are probably more vague in their guidance). Adding explicitly why examining the checklists along different aspects of the SEIPS model will improve the paper.

Thank you for this comment and we appreciate the opportunity to clarify. Our use of The Systems Engineering Initiative for Patient Safety 2.0 (SEIPS 2.0)9 to categorize clinical diagnosis checklists is a novel approach as there is no current framework to examine elements within checklists. Since the literature shows that cognitive, systems and patient factors are root causes for diagnostic error, we used SEIPS 2.0 as it examines each of these factors without placing emphasis on any individual factor. Thus, the tool provides an overall assessment of the landscape of diagnostic error checklists. SEIPS 2.0 is also widely adopted in patient safety and clinical quality improvement research.10-12 We further clarified why we selected SEIPS 2.0 to study checklists in the Introduction, which now reads: “Challenges within cognitive, systems, and patient factors have been identified as important root causes for diagnostic error.23-26 One may hypothesize that checklists may reduce diagnostic error by targeting all of these root causes; conversely those that focus on one or none of these issues are less likely to be successful. Alternatively, a single component of these checklists may represent the “active ingredient” and therefore may be a high-yield target for future research. To date, this empiric question has not been answered. Therefore, in this systematic review, we examined checklists aimed at decreasing diagnostic errors by utilizing an established human factors framework: the Systems Engineering Initiative for Patient Safety 2.0 (SEIPS 2.0).27 The SEIPS 2.0 model has been widely adopted in healthcare research to advance patient safety and quality improvement.28-30 We utilized SEIPS 2.0 to study available clinical diagnosis checklists and understand which human factors are more associated with reducing diagnostic error.” (lines 178-191).

Reviewer 2

10- I would like to see more detail in the Method section regarding the assessment of "risk of bias" to explain what this is, what it involves, and the categorisation used (e.g. good / poor quality), and a more detailed discussion in the Results section and what it means for this particular study. Please see response to item 5 above.
11- There is a "PRISMA 2020 for Abstracts Checklist" attached on the last page of this manuscript. This isn't referred to within the paper and it is unclear what purpose it serves. Is this the completed PRISMA checklist for this paper? If so, it does not make sense that the "risk of bias" under the "Method" section is reported as "No", because this paper does cover risk of bias.

PRISMA is an evidence-based checklist used to outline items for reporting in a systematic review (http://www.prisma-statement.org/). As such, it is not a part of the study itself and is not referenced withing the manuscript text. Rather, it is a guideline that provides clarity on what constitutes a high quality systematic review. As it is a tool to help improve and standardize the reporting in these studies and is typically requested by journals such as BMJ Open, we have provided it here.
PRISMA offers one checklist for the abstract section as well as one for the main body of the text. Within the "PRISMA 2020 for Abstracts Checklist" we answered “No” to item 5 because although we performed a risk of bias assessment in the study, we did not outline the specific methods used in the Abstract due to word count limitations.

Due to the changes made to the abstract and the main text during the review process, we have updated both PRISMA checklists for our systematic review.

VERSION 2 – REVIEW

REVIEWER	Ashley Meyer Baylor College of Medicine
REVIEW RETURNED	11-Apr-2022

GENERAL COMMENTS	The authors have nicely and thoroughly addressed the concerns presented by me and the other reviewers. I have no additional concerns and look forward to seeing this published.